



Figure 1: Illustration of our optimal flow transport and difference to vanilla OT. **(a):** Vanilla OT focuses on solving the transportation within a bipartite graph, focusing on transportation between source and target nodes. **(b):** Our OFT aims to solve transportation on a more general graph, focusing on the flow between each pair of nodes. **(c):** In our OFT, we consider the constraints on each node satisfying the flow balance constraint.

the generalized form of Sobolev transport(ST). However, previous works have mostly overlooked the flow balance constraints on nodes in graph transport. They indirectly compute the flow between two nodes by calculating the shortest paths between source and target nodes, thus failing to incorporate capacity constraints on nodes and edges, which are common in real-world transportation scenarios.

In this paper, we aim to fill this gap by proposing Optimal Flow Transport (OFT), in which the constraints for marginals are replaced by flow balance constraints on nodes. As illustrated in Figure 1, we consider more complex transportation problems where the source nodes (blue) may not directly transport to the target nodes (orange). Instead, it may pass through other nodes, such as transshipment points (yellow). As a result, the original transportation problem is generalized into a broader min-cost flow (MCF) problem where the marginal constraints are replaced by flow balance constraints as shown in Figure 1(c). However, in the current works, we are unable to derive a matrix-vector iterative algorithm for flow balance satisfaction in OFT by introducing an entropic regularization (Benamou et al., 2015) directly. This limitation can be attributed to the following reasons: 1) the marginals are unknown in OFT, and we only have information about their differences; 2) the presence of isolated points in the graph that do not carry any flow contradicts the non-sparse nature of matrix iteration.

To overcome these limitations, in this paper, we propose entropic optimal flow transport, in which we do the following reformulation for the flow problem: 1) we introduce virtual transport flows from each node to itself in the graph, allowing isolated points to participate in the iterations even if they do not have any incoming or outgoing flows originally; 2) reformulating the flow balance constraints with marginal-like constraints via adding a new marginal variable for the optimization; 3) Finally adding the entropic regularization to get the relaxation for OFT. The above reformulations of OFT enable us to iteratively update the coupling and marginals to compute an approximate solution for the OFT problem, where we refer to this algorithm as the OFT-Sinkhorn algorithm, and the theoretical guarantee is also proposed for convergence. Moreover, we can impose capacity constraints on nodes and edges in our OFT-Sinkhorn algorithm, in which the proposed method can serve as a matrix iteration-based solver for the minimum-cost flow problem, and experimental results demonstrate the efficiency of our algorithm with minimal computational errors. Finally, this paper contributes:

- We extend the vanilla OT within a bipartite graph to a more general graph case, and thus propose optimal flow transport, in which the marginal constraints are replaced by flow balance constraints.
- We propose the entropic OFT to derive the GPU-friendly OFT-Sinkhorn algorithm to get the approximate solution of the OFT problem. The global convergence is theoretically guaranteed.
- We incorporate node and edge capacity constraints into the OFT, and in this case our OFT is equivalent to the minimum-cost flow problem. By considering these constraints, we modify the OFT-Sinkhorn algorithm to ensure that the output satisfies capacity constraints. Experimental results on the minimum-cost flow problem showcase the superiority of our algorithm.

2 RELATED WORKS AND BACKGROUND

Entropic Optimal Transport. The Optimal Transport (OT) theory can be traced back to (Monge, 1781) where the objective is to seek a mapping that minimizes the total cost of transporting mass